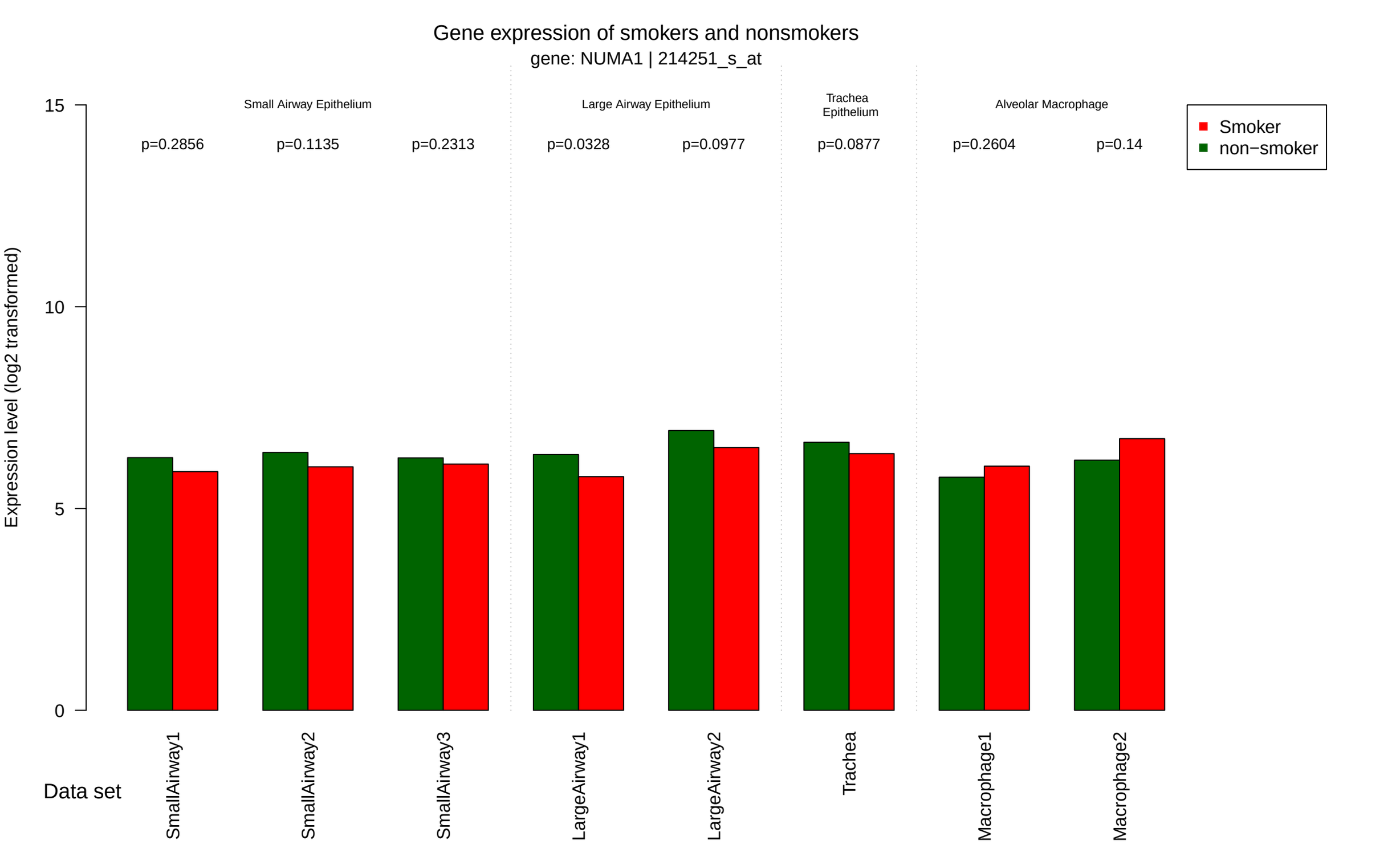
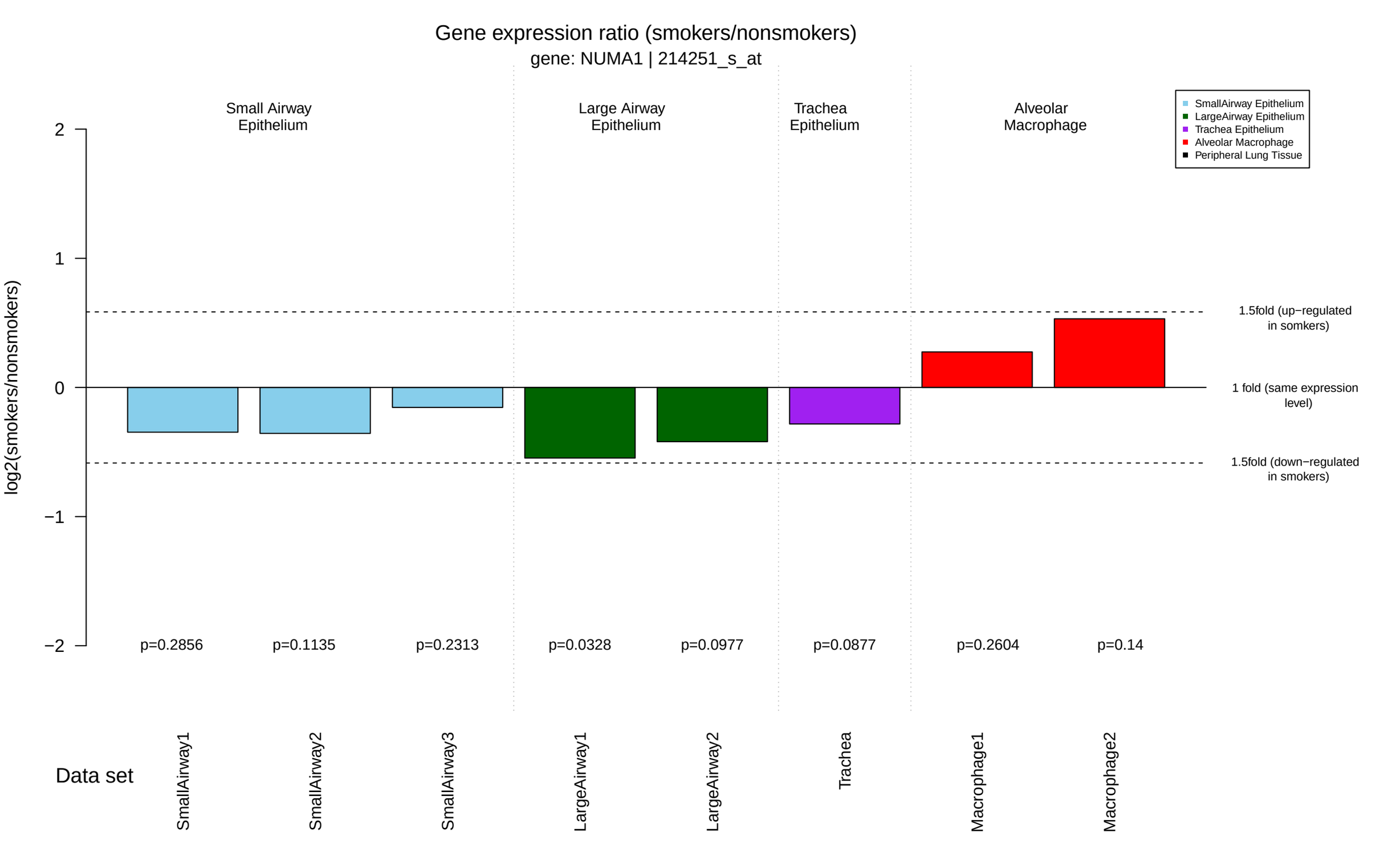
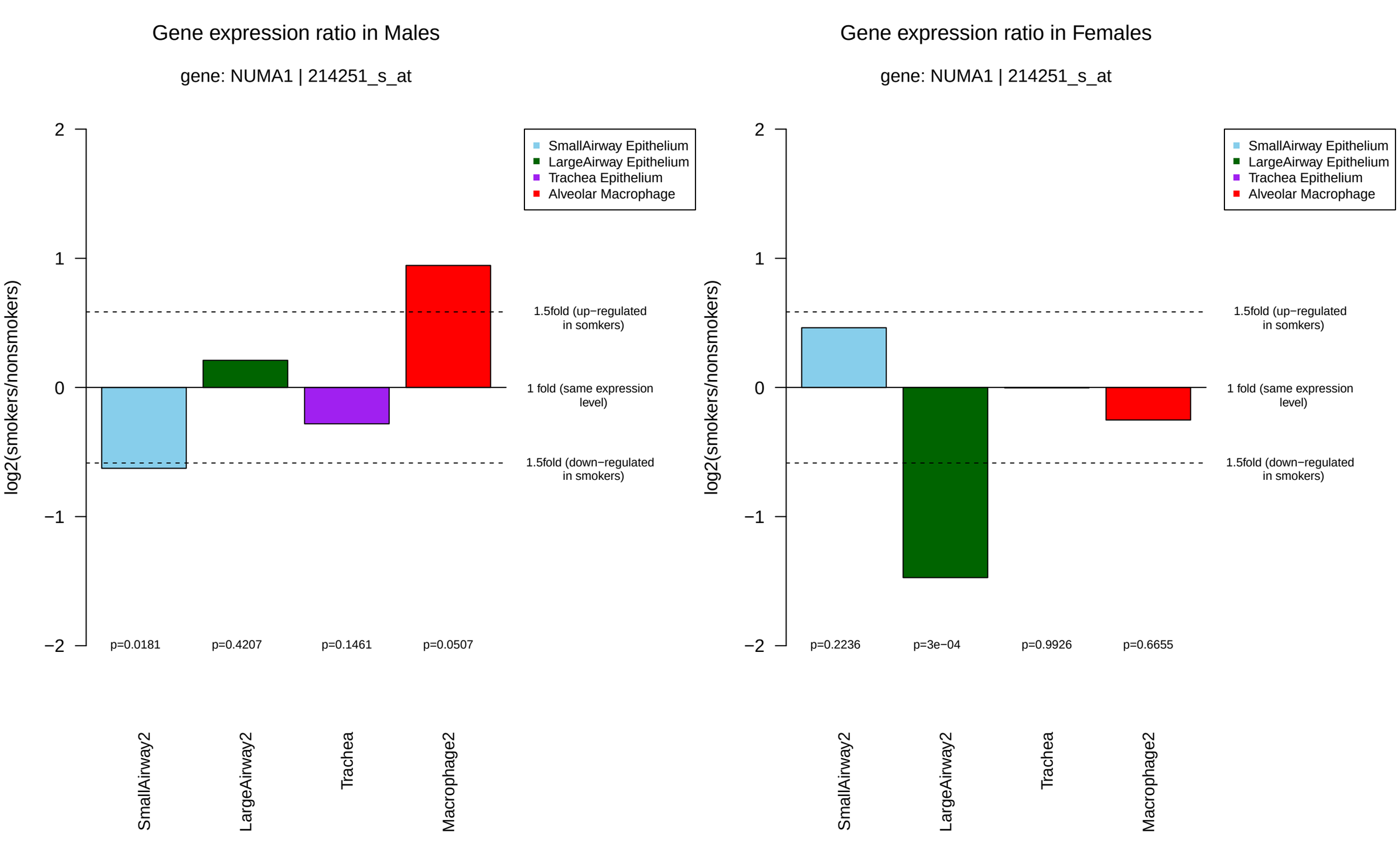
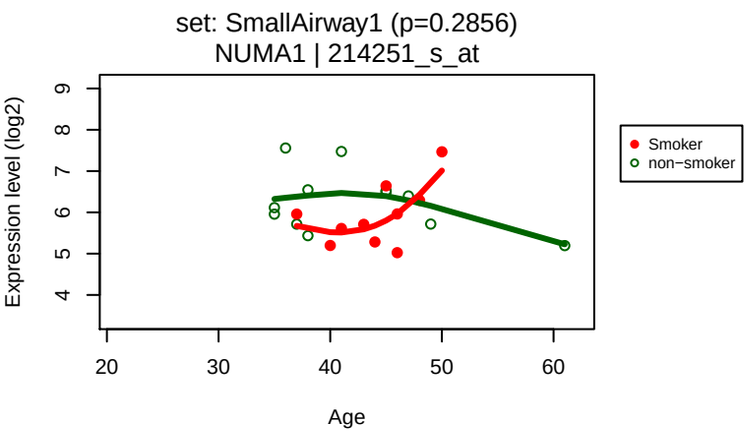
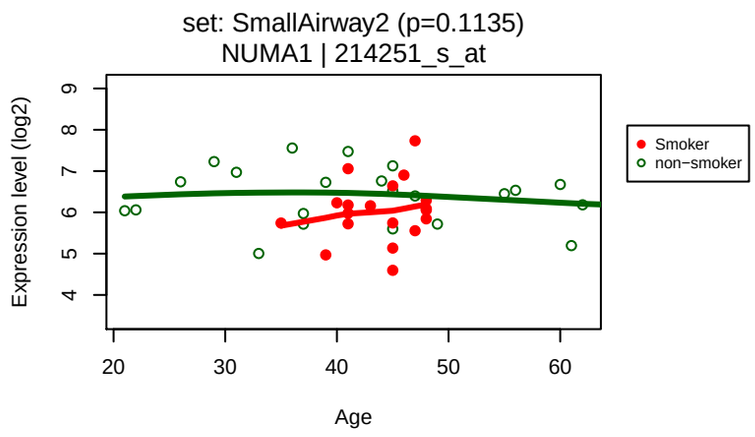
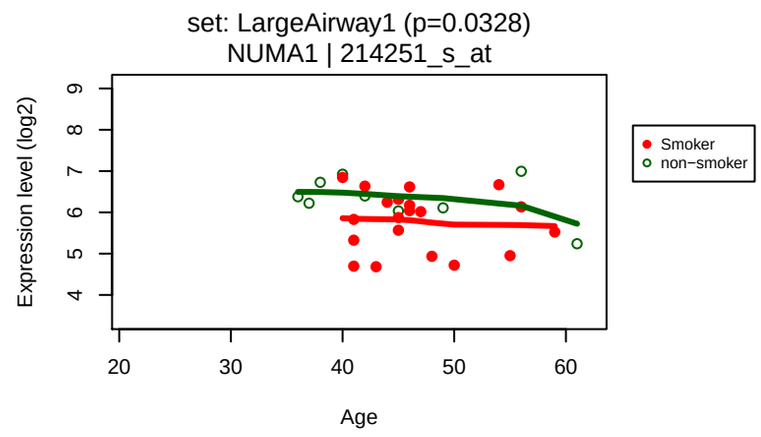

Gene expression of smokers and nonsmokers
gene: NUMA1 | 214251_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2856
p=0.1135
p=0.2313
p=0.0328
p=0.0977
p=0.0877
p=0.2604
p=0.14
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

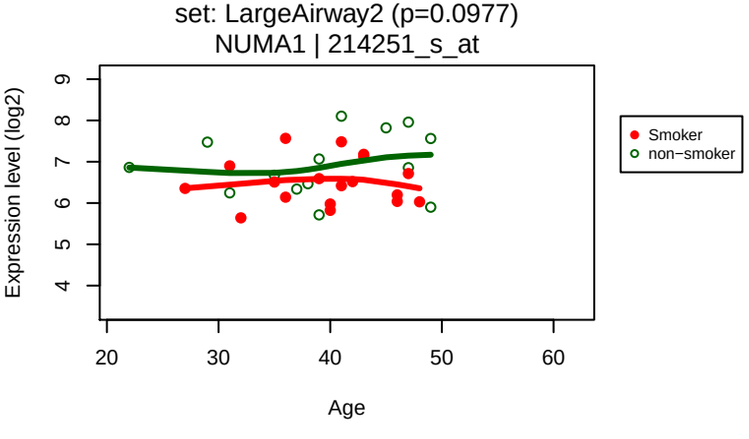

Gene expression ratio (smokers/nonsmokers)
gene: NUMA1 | 214251_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2856
p=0.1135
p=0.2313
p=0.0328
p=0.0977
p=0.0877
p=0.2604
p=0.14
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

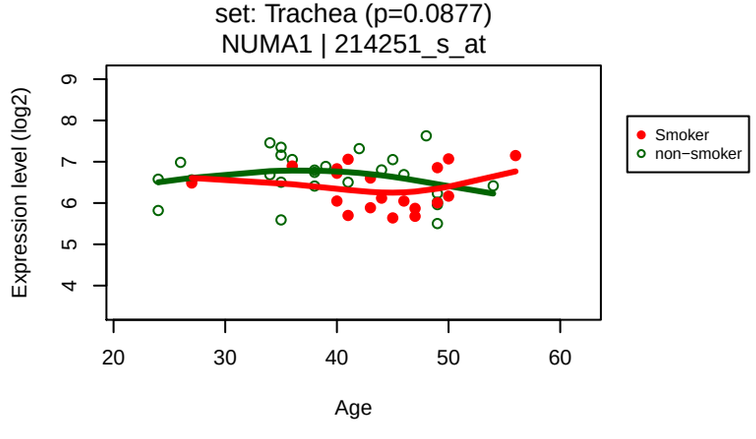

Gene expression ratio in Males
gene: NUMA1 | 214251_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0181
p=0.4207
p=0.1461
p=0.0507
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NUMA1 | 214251_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2236
p=3e−04
p=0.9926
p=0.6655
SmallAirway2
LargeAirway2
Trachea
Macrophage2

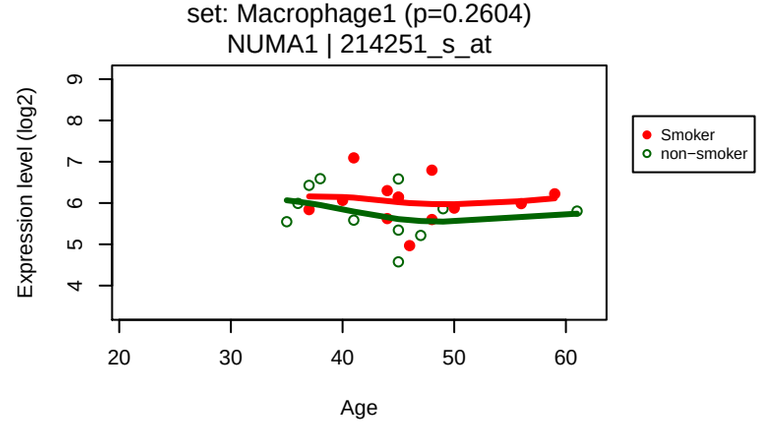

set: SmallAirway1 (p=0.2856)
NUMA1 | 214251_s_at
Expression level (log2)
Age
Smoker
non−smoker

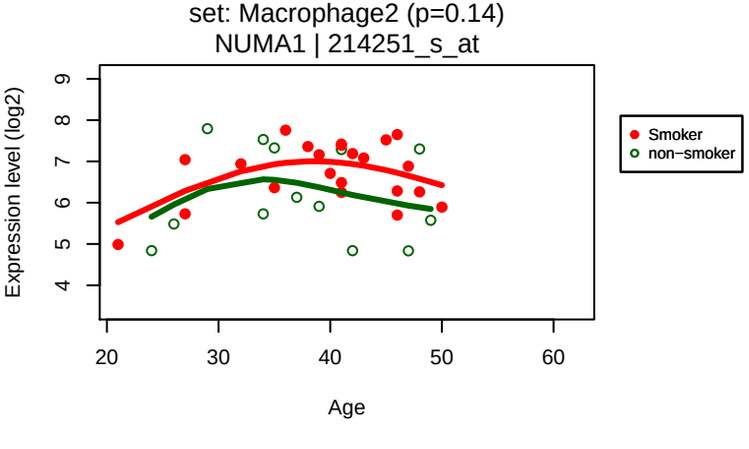

set: SmallAirway2 (p=0.1135)
NUMA1 | 214251_s_at
Expression level (log2)
Age
Smoker
non−smoker

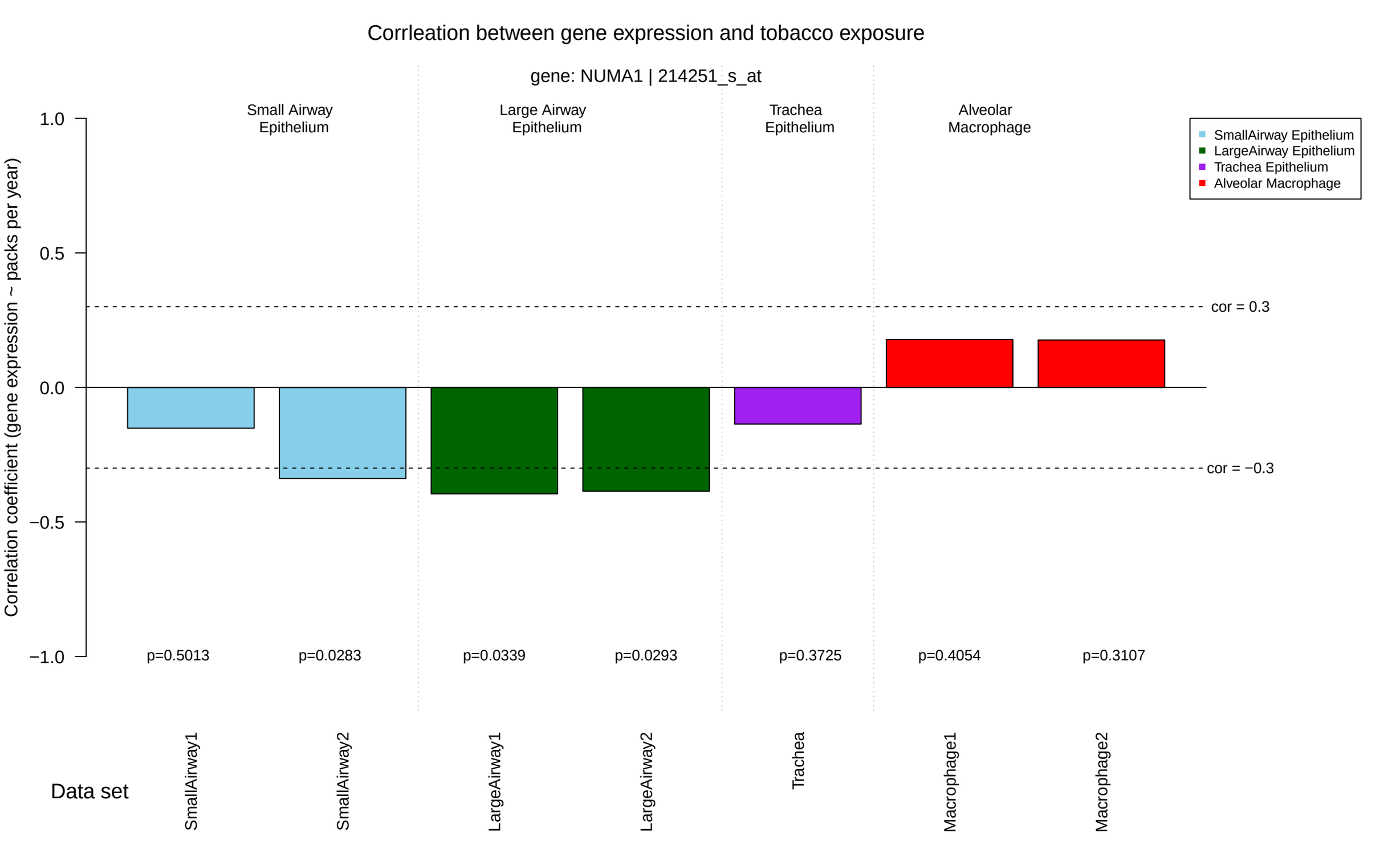

set: LargeAirway1 (p=0.0328)
NUMA1 | 214251_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0977)
NUMA1 | 214251_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0877)
NUMA1 | 214251_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2604)
NUMA1 | 214251_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.14)
NUMA1 | 214251_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NUMA1 | 214251_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5013
p=0.0283
p=0.0339
p=0.0293
p=0.3725
p=0.4054
p=0.3107
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2